ThetaSwap: Adverse Competition Oracle

The Risk Nobody Hedges

Claim: Even if impermanent loss is zero, passive LPs carry residual risk from the *competition structure* of fee revenue [Capponi et al., 2024, Aquilina et al., 2024].

$$\Pi_{\mathrm{LP}} = \frac{\text{Fee Revenue}}{1 + \mathrm{LVC}} - \mathrm{LVR}$$

LVR = Loss Versus Rebalancing (adverse selection) **LVC** = Loss Versus Competition (adverse competition, $\approx \Delta^+$; isomorphic to FLAIR [Milionis et al., 2024])

Risk factor		**Hedging instrument**
IL / LVR	$\longrightarrow$	Variance perpetuals (long volatility) [Fukasawa et al., 2023]; deployed by Panoptic [Kristensen, 2024]
LVC	$\longrightarrow$	**??**

Research Question

Is there empirical demand for a hedging instrument against adverse competition? If so, how much are LPs willing to pay?

Data

ETH/USDC 30bps pool (Uniswap V3 mainnet)
41 days · 600 positions · 3,365 position-day observations · 597 exits

Treatment variable — concentration deviation:

$$\boxed{\Delta^+ = \max\left(0,\ A_T - A_T^{1/N}\right)}$$

- $A_T = \left(\sum_k \theta_k \cdot x_k^2\right)^{1/2}$ fee concentration index (x_k = fee share, $\theta_k = 1/\text{blocklife}$)
- $A_T^{1/N} = \sqrt{\Theta/N^2}$ Ma–Crapis competitive null (equal shares $x_k = 1/N$)
- $\Delta^+ > 0$: fees more concentrated than the symmetric equilibrium

Model: The Shelter Component

$$P(\text{exit}_{i,t}) = \sigma\big(\beta_0 + \beta_1 \boxed{\Delta_{t-\ell}} + \beta_2\, \Delta^2_{t-\ell} + \beta_3\, \text{IL}_t + \beta_4 \log(\text{age}_{i,t})\big)$$

We expect $\beta_1 < 0$ (shelter effect):

- Mild fee concentration correlates with *high trading volume* — the pool is active, all LPs benefit from elevated fee revenue
- At low Δ, more concentration means more volume, which *lowers* exit probability
- This is **healthy competition**: sophisticated LPs attract flow that benefits everyone [Capponi et al., 2024]
- A linear-only model would conclude concentration is *protective* — missing the adverse regime entirely

Model: The Adverse Competition Component

$$P(\text{exit}_{i,t}) = \sigma\big(\beta_0 + \beta_1\,\Delta_{t-\ell} + \beta_2\,\boxed{\Delta^2_{t-\ell}} + \beta_3\,\text{IL}_t + \beta_4\log(\text{age}_{i,t})\big)$$

We expect $\beta_2 > 0$ (adverse competition):

- Beyond a threshold, concentration no longer signals volume — it signals *fee extraction* by sophisticated actors
- The quadratic term Δ^2 captures this **nonlinear regime shift**: the marginal effect of concentration *flips sign* at the turning point
- Together: $\beta_1 < 0 + \beta_2 > 0 =$ **inverted-U** with turning point $\Delta^* = -\beta_1/(2\beta_2)$

Empirical Results

Lag	$\hat{\beta}_1$ (linear)	p	$\hat{\beta}_2$ (quadratic)	p	Turning point
1	$\mathbf{-23.18}$	**0.012**	$\mathbf{+129.20}$	**0.030**	**0.090**
2	-43.42	*0.016*	$+226.92$	*0.065*	*0.096*
3	$\mathbf{-32.44}$	**0.001**	$\mathbf{+205.34}$	**0.004**	**0.079**

For detailed data curation, specification tests, and robustness checks, see `research/model/main.pdf`.

What the Data Says

Passive LPs respond to concentration shocks within one to three days. When fee dispersion exceeds ∼9% above the competitive equilibrium ($\Delta^* = -\beta_1/2\beta_2 = 23.18/(2 \times 129.20) = 0.090$), the shelter effect reverses: each additional percentage point of concentration raises exit probability by approximately 2.3 percentage points ($(\beta_1 + 2\beta_2 \cdot 0.15)\,\bar{P}(1-\bar{P}) = 15.58 \times 0.146 = 2.27$). At a fee dispersion level of 15% above equilibrium, LPs in this pool are willing to pay approximately **10% above the average hourly pool fee revenue** *for protection*
($2.3pp \times 48h$ avg remaining $\approx 1.1h$ of fees, or ∼10% of hourly revenue).

This establishes both the **existence of demand** (the inverted-U with a statistically significant adverse regime) and a **concrete price** (the marginal effect translated to an insurance premium).

Answers

Question	Finding
Is there demand for hedging adverse competition?	**Yes.** LPs react to concentration shocks within days. Below a threshold, concentration is protective; above it, adverse competition drives exits with high statistical significance.
What is the price?	LPs are willing to pay approximately **10% above the average hourly pool fee revenue** for protection — derived from the exit hazard marginal effect at 15% fee dispersion above equilibrium.

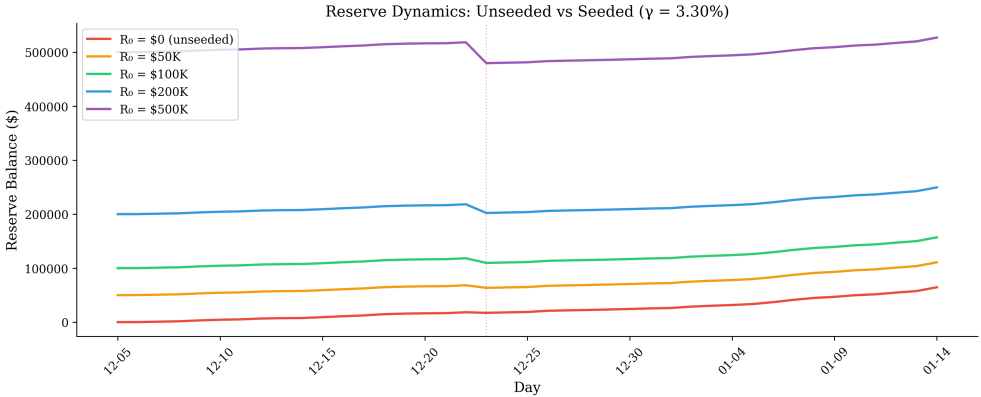
Does Hedging Improve Outcomes?

Historical replay of 600 real positions through a power-squared exit payoff ($\alpha = 2$) with \$200K underwriter seed capital:

Metric	Result
Positions better off hedged	**18.8%**
Mean hedge value	**+\$23.21** per position
Payout / Premium ratio	121.1% (payouts exceed premiums)
Optimal tail exponent	$\alpha = 2$ (dominates $\alpha = 1, 1.5, 3$)

The power-squared payoff rewards LPs who *stay in the pool* during concentration spikes — no moral hazard.

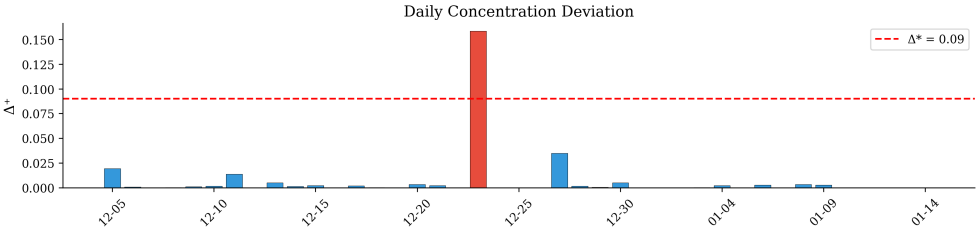
Backtest: Reserve Dynamics



At $R_0 = \$200\text{K}$ seed capital, the reserve absorbs the concentration spike and delivers positive hedge value.

See `notebooks/eth-usdc-backtest.ipynb` and `model/main.pdf` §3 for full methodology.

Tail Risk Confirmation



Only **1 day in 41** exceeds the turning point Δ^*. On that day, Δ^+ jumped from near-zero to **0.158**. **Rare but severe** — a classic insurable tail event.

See `notebooks/eth-usdc-backtest.ipynb` and `model/main.pdf` §5.1 for daily deviation analysis.

FCI Algorithm: Position Tracking

AfterAddLiquidity(*sender*, *key*, *params*)

```
posKey ← Facet.positionKey(sender, params)
Facet.addPositionInRange(posKey, snapshot)
baseline_k ← Facet.poolRangeFeeGrowthInside(poolId, tick, range)
Facet.setFeeGrowthBaseline(poolId, posKey, baseline_k)
N ← N + 1
```

BeforeSwap / AfterSwap(*key*)

```
tick_before ← Facet.currentTick(poolId)
Facet.tstoreTick(tick_before)                          % transient storage
tick_after ← Facet.currentTick(poolId)                 % after swap
Facet.incrementOverlappingRanges(poolId, tick_min, tick_max)
```

All calls are `delegatecall` to the registered protocol facet. Protocols implement `IFCIProtocolFacet` — same algorithm, any AMM.

FCI Algorithm: Index Accumulation

BeforeRemoveLiquidity(*sender*, *key*, *params*)

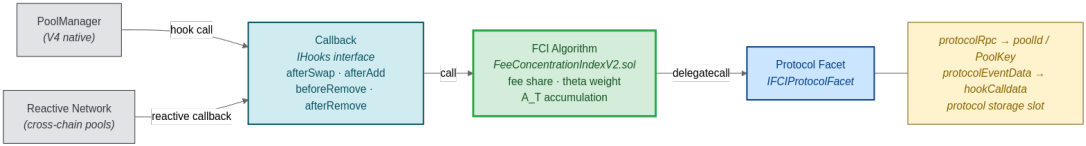
```
(posLiq, feeLast_k) ← Facet.latestFeeGrowth(poolId, posKey)
rangeFee ← Facet.poolRangeFeeGrowthInside(poolId, tick, range)
Facet.tstoreRemovalData(feeLast_k, posLiq, rangeFee)
```

AfterRemoveLiquidity(*sender*, *key*, *params*)

```
if partial remove then return                                   % skip accumulation
(swapLife, blockLife, totalLiq) ← Facet.removePosition(posKey)
if swapLife > 0 then:
    x_k ← fromFeeGrowthDelta(rangeFee, feeLast_k, baseline_k, posLiq, totalLiq)
    θ_k ← 1/blockLife
    A_T^2 += θ_k · x_k^2                                        % state accumulation
N ← N − 1
```

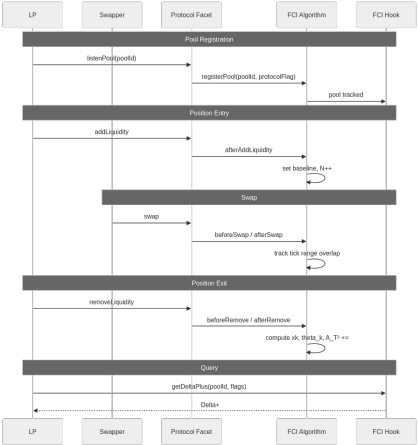
Source: `FeeConcentrationIndexV2.sol`. Query: $\Delta^+ = \max(0, \sqrt{A_T^2} - A_T^{1/N})$ via `getDeltaPlus()`.

Architecture



- **FCI Algorithm** warehouses the logic; facets adapt protocol data via `delegatecall`
- Facet maps: `poolRpc` → `poolId/PoolKey`, `protocolEventData` → `hookCalldata`
- Reactive Network connects pools **on demand** — not limited to V4-native (ref: `protocols/uniswap-v3/`)

Pool Listening Flow



`listenPool()` $\rightarrow$ position entry $\rightarrow$ swaps tracked $\rightarrow$ exit accumulates A_T $\rightarrow$ query Δ^+.

Metrics Interface

What an LP or indexer can query via `FeeConcentrationIndexV2`:

Function	**Returns**
`getIndex(key, flags)`	A_T, Θ, removed position count
`getDeltaPlus(key, flags)`	$\Delta^+ = \max(0, \sqrt{A_T^2} - A_T^{1/N})$
`getAtNull(key, flags)`	$A_T^{1/N} = \sqrt{\Theta/N^2}$
`getThetaSum(key, flags)`	$\Theta = \sum_k \theta_k$
`getDeltaPlusEpoch(key, flags)`	Δ^+ for current epoch window
`getRegistryActiveRanges(key, flags)`	all tracked tick ranges
`getRegistryAllSnapshots(key, flags)`	range-level snapshots (liq, swaps, fees)
`getRegistryPositionBaseline(key, flags, posKey)`	fee growth baseline at entry

Note: Epochs are essential — the cumulative A_T has infinite memory and grows monotonically. Without epoch windowing, the index reflects the entire pool history rather than recent concentration. `getDeltaPlusEpoch()` provides the time-bounded metric.

Interface: `interfaces/IFCIProtocolFacet.sol`

Reference implementation: `protocols/uniswap-v4/NativeUniswapV4Facet.sol`

Live Demo

See it by yourself

```
make sol-test-demo
```

This is a **differential test**: scenarios are generated in Python, expected metric values are computed off-chain, then contrasted against the implied pool state on a Sepolia fork via `forge test`.

What to observe:

1. **Equilibrium scenario** — symmetric LPs, equal fee shares $\Rightarrow \Delta^+ \approx 0$
2. **Crowding-out scenario** — JIT-style position captures disproportionate fees $\Rightarrow \Delta^+$ spikes above Δ^*
3. **Mixed scenario** — combination of passive and sophisticated LPs $\Rightarrow$ shelter regime vs adverse regime in the same pool
4. All metrics computed on-chain, verified against Python-generated expected values

Note: NativeV4 coverage runs on Sepolia fork. 2 tests currently fail due to fixed-point precision cosmetics (rounding in x_k^2 accumulation) — to be resolved.

What's Next

Built

- FCI oracle live on Uniswap V4
- Reactive adapter bridging V3 events cross-chain
- Backtest validation: 18.8% of positions better off hedged, +$23 mean hedge value

Next

- Linearized power-squared CFMM (trading function complete in spec)
- Vault settlement mechanism

The mathematical specification is complete, the oracle tracks metrics across protocols, and the next milestone delivers the full insurance product.

References I

M. Aquilina, S. Foley, L. Gambacorta, and W. Krekel. Decentralised dealers? examining liquidity provision in decentralised exchanges. *BIS Working Papers*, (1227), 2024. Empirical: 7% of LPs capture 75–90% of fees in Uniswap V3.

A. Capponi, R. Jia, and R. Zhu. The economics of just-in-time liquidity. *Working paper*, 2024. JIT LP fee dilution and realized fee share asymmetry.

M. Fukasawa, B. Maire, and M. Wunsch. Weighted variance swaps and impermanent loss. *arXiv preprint*, 2023. Variance perpetuals perfectly hedge IL; `https://arxiv.org/abs/2212.08323`.

J. Kristensen. *Perpetual Options with Uniswap V3: A Practical Guide to Perpetual Options Built on Concentrated Liquidity AMMs*. 2024. Panoptic perpetual options on concentrated liquidity.

J. Milionis, X. Wan, and A. Adams. Flair: A metric for liquidity provider competitiveness in automated market makers. *Working paper*, 2024. Fee-adjusted LP returns metric capturing adverse competition among LPs.